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Paper 1: Characterization of post-marketing requirements for anticancer drugs approved in

Bibliographic Information

- **Title:** Characterization of post-marketing requirements for anticancer drugs approved in Japan: A cross-sectional study
- **Authors:** Tsuno A, Kitagaki H, Maeda H
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Paper Type: GENERIC

Quick Take

This cross-sectional study of 471 anticancer drugs approved in Japan (2001–2024) found that 69.6% received at least one post-marketing requirement (PMR), with risk management plans and all-case surveillance dominating (~89%). Notably, only one drug (oxaliplatin) had a PMR directly related to dose optimization, contrasting sharply with the FDA’s Project Optimus emphasis on pre-approval dose finding.

Executive Summary

The authors systematically characterized PMRs imposed by the PMDA at approval for 471 anticancer drugs approved in Japan between January 2001 and June 2024. They found that 69.6% of drugs received at least one PMR, with the proportion increasing significantly after 2014, coinciding with the introduction of the RMP system in 2013. The most common PMRs were RMP development/implementation (56.1%) and post-marketing all-case surveillance (32.5%), together accounting for ~89% of all PMRs. Only 12 drugs (2.6%) were required to conduct post-marketing clinical trials, all approved between 2001 and 2006, and only one (oxaliplatin) was classified as a dose-optimization PMR. Multivariable logistic regression identified iNDA status, molecular-targeted/immunotherapy mechanisms, hematological tumors, and oral administration as independent predictors of PMR assignment. The authors conclude that Japanese PMRs focus predominantly on safety monitoring rather than dose optimization, highlighting a potential gap in post-approval evidence generation for dose setting.

Scientific Context & Motivation

The paper addresses the regulatory landscape of post-marketing requirements (PMRs) for anti-cancer drugs in Japan, a topic that has been insufficiently characterized compared to the FDA’s approach. The context includes the global shift toward expedited regulatory pathways, the historical ‘higher is better’ MTD paradigm in oncology, and the FDA’s Project Optimus initiative (2021) which emphasizes pre-approval dose optimization. The authors previously examined

dose-optimization PMRs at the FDA (20/138 drugs, 14.5% between 2010-2023) and now aim to characterize the Japanese PMDA landscape, particularly whether dose-related PMRs are used to address dose uncertainty post-approval.

✂ Methodological Snapshot

This is a cross-sectional, retrospective regulatory database study. The authors systematically collected data from PMDA public review reports for all anticancer drugs approved in Japan between January 2001 and June 2024 (new active substances or new therapeutic indications). Exclusion criteria included benign tumors, premalignant conditions, palliative care, and stem cell transplant pretreatment. Data extraction covered pharmacological characteristics, regulatory background, application data packages, and PMR content. PMRs related to dose optimization were defined as post-marketing clinical trial requirements that directly required dose evaluation of the approved drug itself, excluding special populations, food effects, DDI, or concomitant drug dosing. Statistical analyses included descriptive statistics, Cochran-Armitage trend test, chi-square tests for bivariate comparisons, and exploratory multivariable logistic regression to identify independent predictors of PMR assignment. The study followed STROBE guidelines for cross-sectional studies.

📋 Detailed Analysis

The study's most striking finding is the near-absence of dose-optimization PMRs in Japan (0.2%), which contrasts with the FDA's 14.5% rate (20/138 drugs, 2010-2023) reported in the authors' previous work. This difference likely reflects fundamental differences in regulatory philosophy and development strategies. In Japan, the PMDA has historically relied on bridging studies and, more recently, multi-regional clinical trials (MRCTs) to incorporate Japanese patients into global development programs, reducing the need for Japan-specific post-marketing dose studies. The disappearance of post-marketing clinical trial PMRs after 2006 aligns with the increasing use of MRCTs and the ICH E17 framework. However, this raises a critical question: does the absence of dose-optimization PMRs indicate that doses are adequately optimized pre-approval, or does it reflect a gap in post-approval dose evaluation? The paper does not address this directly, but the high proportion of RMPs (56.1%) suggests that safety monitoring is prioritized over dose-related evidence generation. From a pharmacometrics perspective, the reliance on RMPs and all-case surveillance may not generate the quantitative dose-response data needed for model-based dose optimization. The multivariable analysis provides useful insights: iNDA status (OR 6.51) and immunotherapy drugs (OR 22.10) are strong predictors of PMR assignment, reflecting the greater uncertainty and novel toxicity profiles of these agents. The inverse association with solid tumors (OR 0.44) and non-oral routes (OR 0.34) may reflect differences in evidence maturity and risk perception. However, the exploratory nature of the regression and the lack of adjustment for approval year as a continuous variable are limitations. The study's reliance on public review reports introduces potential information bias, as the level of detail varies across drugs and periods. The authors appropriately acknowledge this, but it remains a fundamental limitation. The paper would benefit from a validation of the PMR classification against PMDA internal data or company submissions, though such data are not publicly available. Overall, this study provides a valuable

regulatory baseline and highlights a potential area for policy discussion: whether Japan should adopt a more structured approach to post-approval dose evaluation, similar to the FDA's Project Optimus principles, or whether the current pre-approval focus is sufficient.

Domain Context

This paper sits at the intersection of regulatory science, clinical pharmacology, and oncology drug development. It addresses the global challenge of balancing expedited drug approval with robust evidence generation, particularly for dose optimization. The FDA's Project Optimus (2021) and the 2024 final guidance on dose optimization have shifted the paradigm from MTD-based to exposure-response-based dose selection, emphasizing pre-approval randomized dose-ranging trials. In contrast, the Japanese PMDA has historically relied on post-marketing safety monitoring (RMPs, all-case surveillance) rather than dose-related PMRs. This study quantifies that difference and provides a framework for international regulatory comparison. For pharmacometricians, the paper underscores the importance of pre-approval model-informed drug development (MIDD) to ensure that doses are optimized before approval, as post-marketing dose evaluation is rare in Japan. The findings also have implications for global drug development: if Japanese regulators expect dose optimization to be resolved pre-approval, sponsors must design robust dose-finding studies that include Japanese patients, either through MRCTs or dedicated Japanese studies. The paper's temporal analysis (2001-2024) captures the evolution from bridging-study-based development to MRCT-based development, reflecting the broader globalization of oncology drug development.

☒ Key Findings

- 1) 69.6% (328/471) of anticancer drugs approved in Japan received at least one PMR, with the proportion increasing significantly over time (43.3% before 2014 vs 80.9% during 2014-2024).
 - 2) RMP development/implementation (56.1%) and post-marketing all-case surveillance (32.5%) dominated, representing ~89% of all PMRs.
 - 3) Only 12 drugs (2.6%) required post-marketing clinical trials, all approved between 2001-2006, and only one (oxaliplatin) was classified as a dose-optimization PMR (0.2%).
 - 4) Independent predictors of PMR assignment included iNDA status (adjusted OR 6.51), molecular-targeted drugs (OR 7.76), immunotherapy drugs (OR 22.10), hematological tumors (OR 0.44 for solid vs hematological), and oral administration (OR 0.34 for non-oral vs oral).
 - 5) The findings suggest Japanese PMRs focus on safety monitoring rather than dose optimization, contrasting with the FDA's Project Optimus direction.
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Strengths & Limitations

Strengths

- Comprehensive coverage of a large cohort (471 drugs) over a 23.5-year period, providing robust temporal trends
- Systematic data extraction from PMDA public review reports with independent verification by two investigators

- Clear and predefined classification criteria for dose-optimization PMRs, consistent with prior FDA-focused work
- Use of multivariable logistic regression to identify independent predictors of PMR assignment
- Follows STROBE guidelines for cross-sectional studies, enhancing methodological rigor
- Direct comparison with FDA landscape (previous work) provides international regulatory context

Limitations (Acknowledged by Authors)

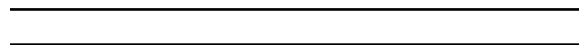
- Reliance solely on publicly available PMDA review reports, which may have omissions or variations in detail
- No systematic evaluation of PMR completion status or outcomes
- No systematic classification of RMP subtypes or assessment of their practical impact
- The association between RMP introduction and increased PMRs should not be interpreted as causal
- Interpretation of the decline in post-marketing clinical trials after 2006 is preliminary, as PMDA policy changes were not systematically evaluated

Limitations (Expert Review)

- The multivariable logistic regression was exploratory and may not have accounted for all confounders (e.g., approval year as a continuous variable, therapeutic area specifics)
- The classification of ‘dose-optimization’ PMRs is narrow and may miss dose-related elements in RMPs or all-case surveillance that could inform dosing
- The study does not assess whether the absence of dose-optimization PMRs reflects adequate pre-approval dose finding or a regulatory gap
- No analysis of the clinical impact or outcomes of PMRs, limiting the ability to judge whether the current system is effective
- The bivariate and multivariable analyses treat each drug as independent, but multiple approvals for the same drug (e.g., imatinib, S-1) may introduce clustering effects

Generalizability

The findings are specific to the Japanese regulatory context (PMDA) and may not generalize to other regions. However, the methodological framework and the observation that dose-optimization PMRs are rare could inform international comparisons, particularly with the FDA and EMA. The temporal trends and predictors of PMR assignment may reflect broader global shifts in oncology drug development, but the regulatory mechanisms (RMP, all-case surveillance) are Japan-specific.



Figures & Tables

- **Figure 1:** Flow diagram showing the selection of 471 anticancer drugs from 2408 approved drugs in Japan (2001-2024), with exclusions for benign conditions, palliative care, and stem cell transplant pretreatment.

- *Significance:* Establishes the study cohort and demonstrates the systematic selection process, ensuring transparency and reproducibility.
- **Figure 2:** Temporal trend of the number of drugs requiring PMRs versus those without PMRs, showing a crossover around 2014 and a significant increasing trend (Cochran-Armitage $Z = 5.760$, $p < 0.001$).
 - *Significance:* Visualizes the key temporal finding that PMR assignment has become more common, likely influenced by the 2013 RMP system introduction.
- **Table 1:** Characteristics of 471 oncology drugs approved in Japan, including NDA type, mode of action, tumor type, special regulatory pathways, development style, companion diagnosis, and route of administration.
 - *Significance:* Provides the descriptive baseline for the cohort, essential for interpreting the regression analyses.
- **Table 2:** Distribution of PMR counts per drug, showing 69.6% with at least one PMR and the majority (46.9%) with a single PMR.
 - *Significance:* Quantifies the overall PMR burden and shows that multiple PMRs are relatively uncommon.
- **Table 3:** Breakdown of PMR contents, with RMP (56.1%) and all-case surveillance (32.5%) dominating, and post-marketing clinical trials representing only 2.6%.
 - *Significance:* Central finding that Japanese PMRs are safety-focused rather than evidence-generation-focused for dose optimization.
- **Table 4:** List of 12 drugs requiring post-marketing clinical trials (2001-2006), with reasons and development style (bridging/MRCT).
 - *Significance:* Demonstrates that post-marketing clinical trials were historically used to address ethnic/regional evidence gaps, not dose optimization, and have disappeared since 2006.
- **Table 5:** Bivariate associations between drug characteristics and PMR assignment, showing significant associations with NDA type, mode of action, tumor type, public knowledge-based application, MRCT, companion diagnosis, and route of administration.
 - *Significance:* Provides the basis for the multivariable model and highlights which factors are associated with PMR assignment.

Code & Reproducibility Assessment

The study used Microsoft Excel 2019 and R version 4.5.2 for analyses. No code or data repository is mentioned in the abstract or main text. The database was constructed from public PMDA review reports, but the customized database is not publicly available. Reproducibility would require re-extraction from PMDA sources.

Supplementary Materials

Supplementary materials include Figure S1 (distribution of tumor types), Figure S2 (temporal trends in approvals), Table S1 (detailed subcategories of PMRs including medical institution restrictions), Table S2 (classification of dose-optimization PMRs), and Table S3 (multivariable logistic

regression results). These provide additional detail on the classification criteria and regression outputs.

Future Directions

1) Systematic evaluation of PMR completion status and outcomes in Japan, including whether RMP activities generate meaningful safety data. 2) Detailed disaggregation of RMP subtypes and assessment of their clinical impact. 3) Comparison of Japanese PMR practices with FDA and EMA, particularly regarding dose-optimization requirements. 4) Investigation of whether the absence of dose-optimization PMRs correlates with adequate pre-approval dose finding or with post-approval dose changes (label changes). 5) Analysis of the impact of ICH E17 and multi-regional clinical trials on the need for Japan-specific post-marketing studies. 6) Prospective tracking of dose-related label changes post-approval to assess whether the current system adequately addresses dose uncertainty.

Expert Commentary

This paper provides a valuable regulatory epidemiology perspective that complements the pharmacometrics community's focus on model-informed drug development. The finding that only 0.2% of Japanese anticancer drugs have dose-optimization PMRs is striking, especially given the FDA's Project Optimus emphasis on pre-approval dose finding. From a pharmacometrics standpoint, this suggests that dose optimization in Japan is expected to be achieved through pre-approval PK/PD modeling and dose-ranging trials, rather than post-marketing studies. However, the high proportion of RMPs and all-case surveillance (89%) indicates a heavy reliance on post-marketing safety monitoring, which may not generate the quantitative dose-response data needed for model-based dose optimization. The paper's limitation in not assessing PMR outcomes is important—future work should link PMR data with label changes and dose adjustments to determine whether the current system effectively supports evidence-based dosing. The multivariable analysis identifying immunotherapy drugs as having the highest odds of PMR assignment (OR 22.10) is particularly relevant given the rapid expansion of immuno-oncology agents and their unique toxicity profiles.

Bottom Line

For pharmacometrics and regulatory scientists, this study provides a comprehensive baseline of Japanese PMR practices, showing that post-marketing requirements are overwhelmingly safety-focused (RMPs and all-case surveillance) rather than dose-optimization-focused. The near-absence of dose-related PMRs (0.2%) suggests that dose optimization is expected to be resolved pre-approval, aligning with the global shift toward Project Optimus principles. However, the study also highlights a potential evidence gap: if dose uncertainty exists at approval, the current Japanese PMR system may not systematically capture it. This underscores the need for robust pre-approval dose-finding studies and possibly a more structured post-approval dose evaluation framework.

Fact-check corrections